Independent study — report through Week 8

physio-placebo

Is the model reading the recording, or the words we put around it?

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This edition	14 October 2026
Closed through	Week 8
Still ahead	Week 9, then the summary score in Week 10
Repository	`https://github.com/shrijacked/physio-placebo`
Pre-registration	`https://osf.io/62r5t` DOI 10.17605/OSF.IO/62R5T

Every score uses macro-F1. For two classes, that is the average of how well the model catches each class, so always guessing the common class cannot look brilliant. On these datasets that guessing score is about 0.41, not 0.50. A bracket is a 95% interval from resampling people, not windows. Seed 1337 throughout.

1 Where things stand

Block	State	One-line result
Week 1 — data	Closed	Three public datasets, windowed, no windows dropped
Week 2 — simple models	Closed	On WESAD, logistic regression reaches 0.92
Week 3 — a published number	Closed	We did not reproduce Health-LLM's 0.76
Weeks 4–6 — real recordings	Closed	Best language-model score is 0.60; plots are at chance
Your comparison request	Done	Prompting and fine-tuning kept in separate tables
Weeks 7–8 — replacements	Closed	Noise collapses WESAD; spectrum and "same condition" do not
Week 9 — adjectives	Not started	Strip the words, then flip them
Week 10 — one summary ratio	Not started	Only after Week 9

The short version. On WESAD, logistic regression on ordinary heart and skin features reaches 0.92. The best language model on the real recording reaches 0.60. Showing the model a plot instead of numbers does nothing: it sits at chance. When I replace the recording, Gaussian noise, a flat line, an empty prompt, and another person in the wrong condition all fall to that same chance line. Two replacements do not. A phase-randomized ECG, which keeps the spectrum, still scores 0.88 with logistic regression. Another person who was in the same condition scores 0.95. The language model, when it is reading feature sentences, follows that pattern and even exaggerates it.

2 The study, in one paragraph

A run of papers now claims that a language model can tell stress or mental workload from a wearable. They rarely check whether the model needed the recording. I take three public lab datasets, show each 60-second window to an open model in three ways, and then replace the recording with noise, with another person, or with nothing. If the score survives, the model

was not reading that trace. The sharper test, still to come, is whether it is reading the numbers or the adjectives around them ("elevated heart rate, stress-typical"). Either outcome is worth reporting. That was filed before these runs, at `https://osf.io/62r5t`.

The model does not write a paragraph. For each window I look at the probability of two letters. `A` is the calmer class (non-stress, low workload, or rest). `B` is the harder class (stress, high workload, or Stroop). Whichever letter is more probable is the prediction. The wording of the prompt was chosen on two people and then left alone. Those two people are not in the scores below. For the verbalized features, the words "high" and "low" are set from the other people, never from the person being scored. There is no training of the language model on WESAD, MAUS, or CogWear. Fine-tuning appears only in the Health-LLM comparison, because that paper fine-tuned.

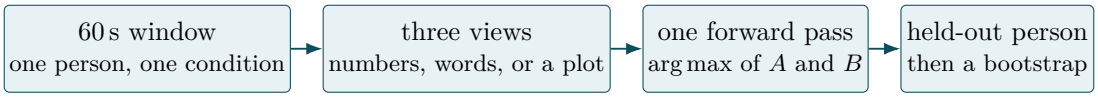


Figure 1: What one scored window is. The prompt wording is chosen on two people and then frozen. The numbers in this report are the other people.

Here is one WESAD window, shown the three ways the models see it. **A** is the ECG itself, averaged into 128 numbers so it fits in a prompt. **B** is eighteen short sentences about heart rate, heart-rate variability, skin conductance, and breathing. Four of them are printed below. Words like "stress-typical" are deliberate. Week 9 asks what happens when they are removed. **C** is a line plot of the ECG, with no name and no class written on it. The plot in Figure 2 is this kind of picture. Figure 3 is a different window, subject S10 during stress, drawn as a trace so the later replacements are easy to compare by eye.

Numbers — the ECG averaged into 128 bins

```
ecg fs=700 Hz n=128 scheme=bin_mean:
0.04024 0.005173 -0.01352 -0.01912 0.03245 -0.00422 -0.02569 0.0196
... 117 values omitted ...
-0.04705 -0.00251 -0.06615
```

Sentences — 4 of the 18 feature lines

```
eda_tonic_mean is reduced (0.7358), calmer-typical.
hrv_mean_hr is elevated (89.28), stress-typical.
hrv_sdnn is reduced (41.18), stress-typical.
hrv_rmssd is reduced (17.7), stress-typical.
```

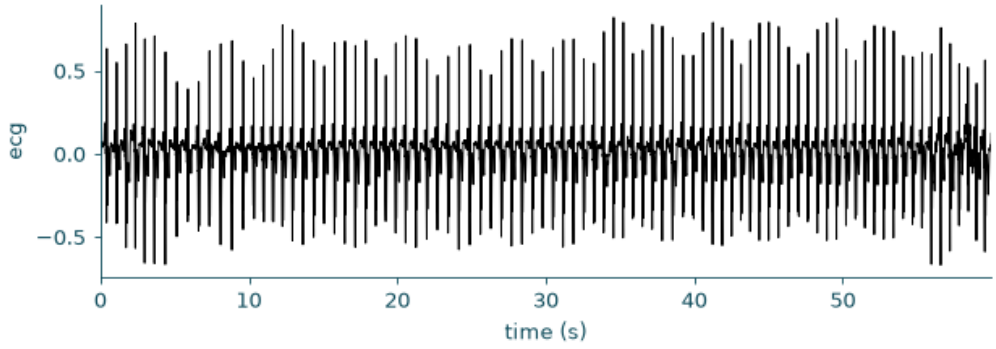


Figure 2: The plot the vision model sees for the window quoted above. No name, no class.

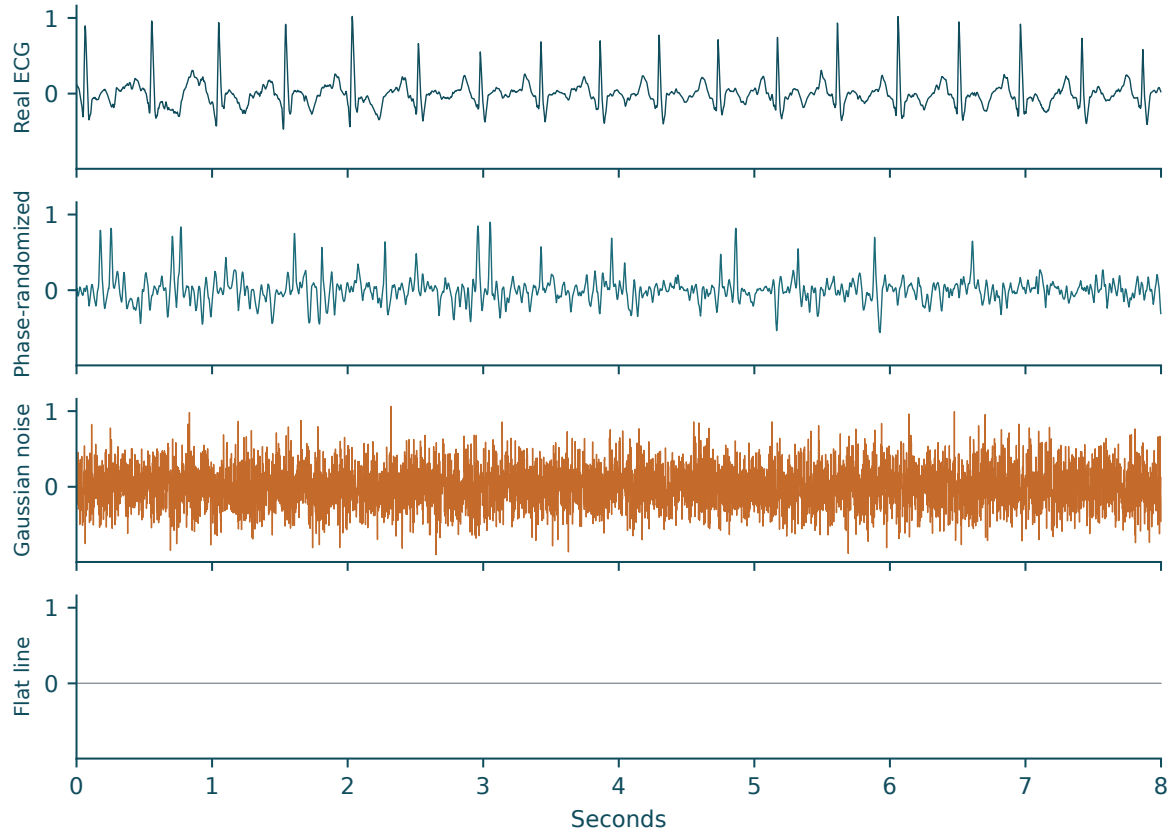


Figure 3: Eight seconds from one stress window of WESAD subject S10, chest ECG. The top trace is the recording. The three below it are the replacements used later: phase-randomized, Gaussian noise matched to the whole dataset, and a flat line at that same mean. The person and the label stay put. Only the trace changes.

3 The three datasets

Each example is one minute of a recording, and the next window starts 30 seconds later, so neighbors overlap. A window never crosses from one condition into another. WESAD is the standard lab stress set: a public speaking task against calm and a comedy clip, from a chest belt and a wristband, 15 people. MAUS is easy versus hard n-back, from a wrist pulse, 22 people. CogWear is a color-word Stroop against rest, from a wristband, 10 people. None of the windows were dropped for missing samples.

Dataset	People	Windows	Class split	Dropped
WESAD	15 / 15	1,042	731 non-stress / 311 stress	0
MAUS	22 / 22	1,086	362 low / 724 high	0
CogWear	10 / 11	114	39 baseline / 75 Stroop	0

CogWear subject 3 is out because PhysioNet is missing that person's EDA file. The feature code is frozen (NeuroKit2 0.2.13). The same code feeds the linear model and the feature sentences, so those two are looking at the same numbers.

4 What a simple model can already do

Before any language model ran, I fit a logistic regression and a small decision tree on the same features the sentences are built from. The shuffle ceiling is a sanity check: shuffle the labels

inside each person, refit a thousand times, and take the score that 95% of those shuffles fall below. A real model has to clear that line. "Guessing" is always picking the more common class.

Dataset	Logistic reg.	Small tree	Guessing	Shuffle ceiling	Signal
WESAD (n=15)	0.918	0.848	0.412	0.433	Strong
MAUS (n=22)	0.444	0.482	0.400	0.413	Weak
CogWear (n=10)	0.569	0.560	0.397	0.541	Weak

WESAD is the dataset that can test whether a model is using the recording. A linear model is far above chance there, so an LLM that ignores the trace has a high score to miss. MAUS and CogWear are weak even for logistic regression. On those two, a language-model score near 0.4 is not evidence of anything, and the ratio we pre-registered for later will often be undefined. The raw gap is what I will report.



Figure 4: Best simple model against the best language-model score on the real recording. Each language-model bar is the best of that model's three settings. WESAD is the only dataset where the simple model is far above guessing.

5 Checking one published number before trusting the pipeline

The plan was filed publicly before the language-model grid ran (`https://osf.io/62r5t`). Three predictions are on that filing, and none of them are scored yet. I expect plots and raw number-lists to lose most of their advantage once the recording is replaced. I expect the worded features to lose more than half their advantage once the adjectives are stripped. I expect flipped adjectives ("low" where the value is high) to change the prediction on more than 40% of windows. Week 10 is where those get a single number. Week 9 is the adjective half.

As a check that I was running the published method and not a lookalike, I reproduced one Health-LLM setting before the rest of the grid: PMData stress, MedAlpaca-7B, no examples in the prompt. Their paper says MAE 0.76 ± 0.1. This rerun is $\mathbf{2.689 \pm 0.038}$. The answers are often "out of 5" or "out of 10", and the scorer, which was fixed in advance, reads the first number, so it records 5 or 10. I left that rule alone. The gap is part of the result, not a bug I papered over.

The models in the rest of this report are open weights: Qwen3-8B and Llama-3.1-8B for text, and Qwen2.5-VL-7B for plots. The registration originally named a GPT model. That path was dropped because the account was capped at 50 requests a day, and the registration has a public note saying so.

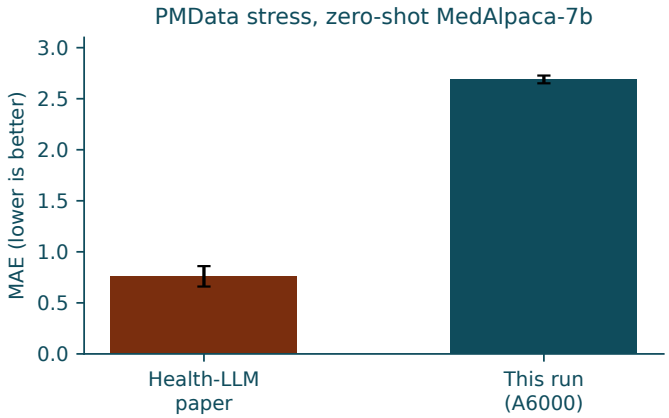


Figure 5: The published Health-LLM number (0.76 ± 0.1) next to this rerun (2.689 ± 0.038). Lower is better. The gap is the result.

6 Language models on the real recording

"Every nth" keeps one sample out of every few, so the minute still fits in a prompt. "128 averages" chops the minute into 128 bins and averages inside each bin. "Sentences" are the feature lines. The two people used to pick the wording are WESAD S2 and S8, MAUS 015 and 023, and CogWear 4 and 8. No window failed to score.

Dataset	What the model sees	Qwen3-8B	Llama-3.1-8B
WESAD	every nth sample	0.427 [0.392, 0.465]	0.333 [0.301, 0.370]
WESAD	128 averages	**0.595 [0.550, 0.647]**	0.230 [0.227, 0.233]
WESAD	feature sentences	0.584 [0.462, 0.699]	0.283 [0.247, 0.324]
MAUS	every nth sample	0.467 [0.436, 0.499]	0.413 [0.403, 0.426]
MAUS	128 averages	0.483 [0.452, 0.516]	0.400 [0.400, 0.400]
MAUS	feature sentences	0.487 [0.439, 0.531]	0.400 [0.400, 0.400]
CogWear	every nth sample	0.576 [0.443, 0.689]	0.404 [0.392, 0.417]
CogWear	128 averages	0.396 [0.380, 0.411]	0.440 [0.399, 0.495]
CogWear	feature sentences	0.475 [0.393, 0.504]	0.479 [0.375, 0.640]

The bold row is the best language-model result in the study so far: Qwen, on WESAD, reading the ECG as 128 averages, with no worked examples in the prompt. It clears guessing (0.41). It does not come close to logistic regression (0.92). Llama's two MAUS rows at 0.400 with no interval width are not a crash. On every held-out person, Llama gave the same answer every time. CogWear's intervals are wide because only 90 windows are scored.

7 Showing the model a plot instead

Qwen2.5-VL-7B sees a line drawing of the chest ECG on WESAD, or the wrist pulse on MAUS. The drawing has no name and no class on it. The same window always makes the same file. CogWear has no plot condition. The prompt here includes four example plots, chosen before these scores.

Dataset	Prompt	macro-F1 [95% CI]	Windows
WESAD	four example plots	0.412 [0.410, 0.414]	906
MAUS	four example plots	0.400 [0.399, 0.400]	990

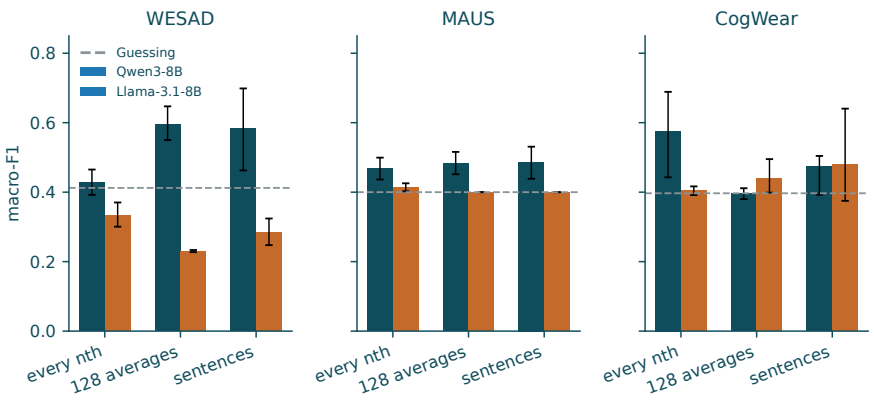


Figure 6: Language models on the real recording. Dark bars are Qwen, orange bars are Llama. The dashed line is guessing. The only bar that clearly clears that line, with the interval above it, is Qwen on WESAD with the averaged ECG.

Both sit on chance, and the intervals are tight because the model almost always picks the same class. That is what we expected for plots. I have not gone back and rewritten the plot prompt to chase a higher number.

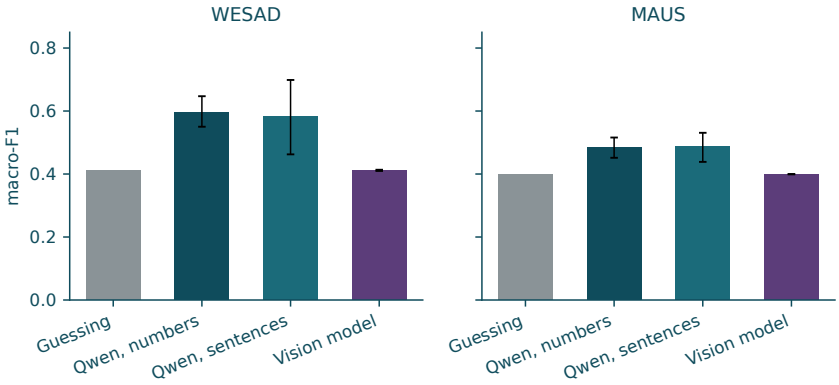


Figure 7: The plot condition next to guessing and next to Qwen's number and word conditions. The vision model does not beat guessing on either dataset.

8 What you asked for: a matched comparison

After the last meeting you asked for recent papers, from a real venue, and for the comparison to stay matched. Same public dataset, same open model, and prompting next to prompting, fine-tuning next to fine-tuning. A WESAD macro-F1 does not belong next to a paper that reports MAE on daily step counts.

The paper that meets that bar is Health-LLM (Kim et al., CHIL 2024). The task is PMData stress, a score from 1 to 5. The metric is mean absolute error: how far the predicted score is from the person's own rating. Lower is better. The inputs are two weeks of steps, calories, resting heart rate, sleep, and that day's mood. The model with public weights is MedAlpaca-7B. Their Table 3 is frozen prompting. Their Table 4 is the same model after fine-tuning. SensorLM (NeurIPS 2025) is a conference paper in the same area, but the weights and the Fitbit corpus are not public, so it is not in these tables.

The zero-shot rerun uses their prompt and the locked 299-item eval. The first-number scorer was fixed before the run. Many completions say "out of 5" or "out of 10", and the scorer reads 5 or 10. That is why our MedAlpaca MAE is far above theirs. I did not change the scorer to

close the gap.

Row	Model	Adaptation	MAE
Paper Table 3	MedAlpaca-7b	no examples	0.76 ± 0.1
This rerun	MedAlpaca-7b	no examples	2.689 ± 0.038
Our model, their prompt	Qwen3-8B	no examples	0.795 ± 0.020
Our model, their prompt	Llama-3.1-8B	no examples	2654 ± 43

Qwen answers `Answer: 3`. Llama writes Python that pastes step counts; the locked first-number scorer reads those counts. The scorer was not changed. Always answering 3 on this eval has MAE 0.401, which is still below Qwen.

Row	Model	Adaptation	MAE
Paper Table 4	MedAlpaca-7b	fine-tuned on PMData	0.38
This rerun	MedAlpaca-7b	fine-tuned, small adapter	2.697

297 of 299 fine-tuned answers parsed. They still say "out of 5", so the scorer still reads 5. The same model's zero-shot seed on this eval was 2.714. Fine-tuning did not move it. Their published fine-tuned error is 0.38. Ours is 2.70. Training follows their 7B recipe: 3 epochs, batch 128, learning rate 2×10^{-5}, and the small adapter from the MedAlpaca defaults. I did not oversample, because they never published the rule, and I skipped the 100-step warmup, because this training set is only 653 rows. Qwen, on their prompt, lands near their paper number and still loses to always answering 3 (error 0.401 on this split). Llama does not answer with a stress score. It writes Python, and the scorer reads a step count out of that code.

9 Weeks 7–8 — what survives when the recording is replaced

The prompts stay frozen. For each window I swap the trace and score again. There are five replacements that are meant to destroy the signal, and one that is not. That last one keeps the condition and changes the person. It answers a different question: is the model using this recording, or the fact that stressed people look alike? It is in the tables. It is not a null.

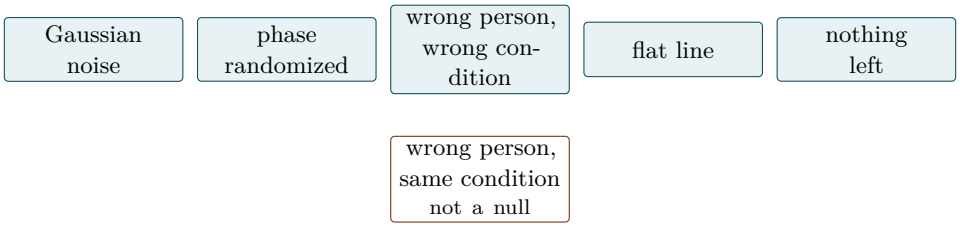


Figure 8: The five replacements that are meant to wipe the class, and the one underneath that keeps it on purpose.

One hundred and twenty settings were scored this way, with no failed windows. The probability of each letter is saved with every window. Before looking at the language models, the same features go through the same logistic regression as in Week 2. That table says how much class information each replacement still holds. A replacement is not assumed to be empty. It is measured.

Classical logistic regression on those features:

Replacement	WESAD	MAUS	CogWear
Gaussian noise	0.412	0.400	0.486
Phase-randomized	0.880	0.405	0.458
Wrong person, wrong condition	0.411	0.405	0.380
Flat line	0.412	—	0.397
Signal deleted	—	—	—
Same condition, other person	0.954	0.494	0.773

A dash means there was nothing to fit. A flat MAUS pulse has no heartbeats, so the features are empty. Deleting the signal leaves no features on any dataset.

Read the WESAD column from the top. Noise, a flat line, and a window from the wrong person in the wrong condition all fall to 0.41, which is the majority floor. A phase-randomized ECG, which keeps the spectrum and throws away the waveform, is still at 0.88. Another stressed person's recording, with the label kept, is at 0.95. So on this dataset the class is mostly in the spectrum and in the kind of person, not in one individual's pulse shape. MAUS and CogWear never had that strong a classical signal, and the ladder does not create one.

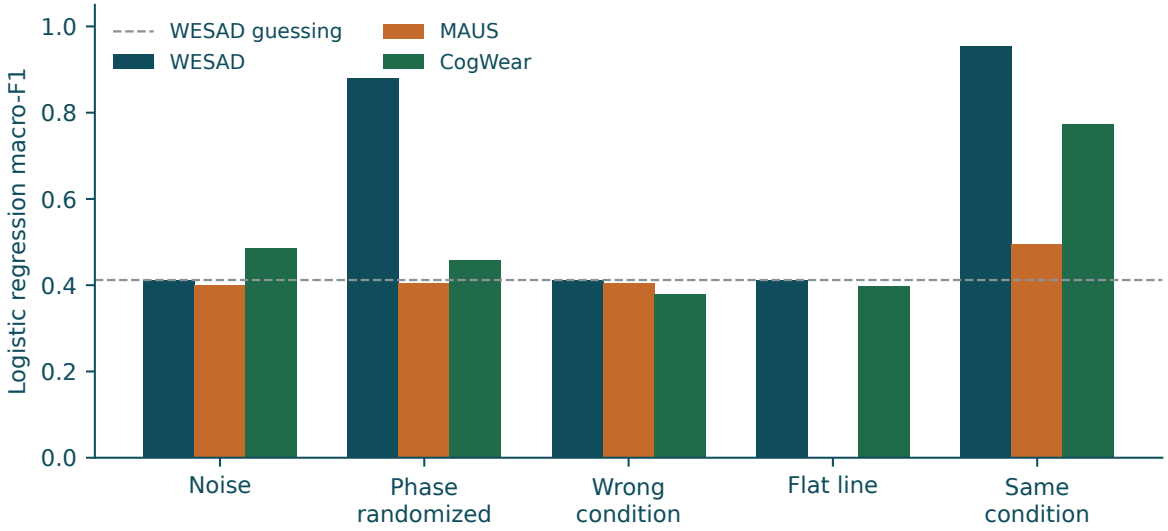


Figure 9: Logistic regression on each replacement. The missing MAUS flat-line bar is the empty-feature case above, not a score of zero. The dashed line is guessing on WESAD.

The same replacements were then given to the frozen prompts. Below is Qwen3-8B on WESAD, the only place a language model had cleared chance on the real recording. Llama's intervals all touch guessing or sit under it, so a second chart would not show a pattern. Green is the real recording. Brown keeps the condition and changes the person. Teal is a null. The legend is on the figure.

That last pair is the result I want on the table. When the words in the prompt are built from features that still contain the class, Qwen does well even if the waveform is no longer the original person's. When the features are noise or absent, Qwen falls to chance or below it. The raw-ECG setting moves less. It was only modestly above chance to begin with, and several replacements stay in that same band. The dependence score that turns these gaps into one ratio is Week 10. It is not in this PDF.

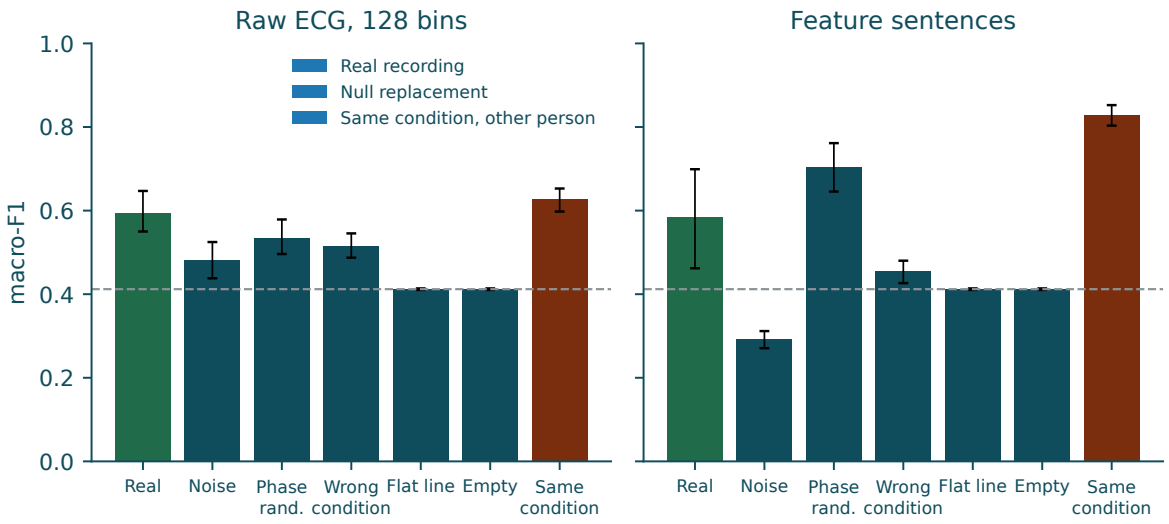


Figure 10: Qwen3-8B on WESAD, held-out people only. Left: ECG as 128 averages (0.60 on the real recording). Right: feature sentences (0.58 on the real recording). Green is the real recording. Teal bars are the null replacements. Brown keeps the stress label and changes the person. A flat line and an empty prompt sit on chance. Sentences built from a phase-randomized trace score 0.70, higher than the real recording. Sentences from another stressed person score 0.83.

10 What is left

Week 9 is the adjective test, on the feature sentences only. Three wordings of the same numbers: as they are now, with the evaluative words removed, and with those words flipped. There is an optional fourth wording, SensorLM's plain statistical sentences (mean, max, min, spread), with none of their activity or mood lines. I would like a yes or no on that fourth wording before I freeze the set.

Week 10 turns the gaps in this report into one ratio per condition,

$$\mathrm{SDS} = \frac{M_{\mathrm{real}} - M_{\mathrm{surrogate}}}{M_{\mathrm{real}} - M_{\mathrm{chance}}},$$

where M is macro-F1 and M_{chance} is the shuffle floor from Week 2. If the real recording was never convincingly above that floor, the ratio is left blank and only the gap is shown. MAUS and CogWear are where that will happen. The ratio is not in this PDF.

Shrijak Kumar • 14 October 2026 • Dr. Siddharth • Plaksha University